HSE 103 Final Project

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1 Methods

1.1 Data Source

The data used in this project were obtained from Gene Expression Omnibus dataset GSE85307, which was generated as part of a study examining early pregnancy vitamin D status and risk of preeclampsia [3]. The provided dataset contained 157 maternal samples. Gene expression measurements were matched to maternal covariate data using GEO sample identifiers.

All 157 samples were retained in the final analytic dataset. No samples were excluded from the primary analyses.

1.2 Data Processing and Analysis

Three genes, *TBL1X*, *KDM1B*, and *TMEM176B*, were selected for the primary gene expression analyses. Probe identifiers were matched to gene names programmatically in R using the provided gene-name mapping file.

Maternal body mass index (BMI), maternal vitamin D concentration, maternal age, maternal asthma status, and pregnancy outcome were cleaned and reformatted prior to analysis. A categorical BMI variable was created by dividing maternal BMI at the sample median into below-median and above-median BMI groups.

For each of the three selected genes, a histogram was used to describe the distribution of gene expression, a scatterplot was used to visualize the relationship between maternal vitamin D concentration and gene expression, and a boxplot was used to compare gene expression across pregnancy outcome and maternal BMI group.

Summary statistics were calculated for maternal age, maternal BMI, maternal vitamin D concentration, maternal asthma status, and maternal BMI group, stratified by pregnancy outcome. Continuous variables were summarized using means and standard deviations. Categorical variables were summarized using counts and percentages.

A clustered heatmap was generated using 10 genes. The three primary genes were retained, and seven additional genes were selected based on highest expression variance. Genes and samples were clustered using Euclidean distance. Maternal BMI group and pregnancy outcome were displayed as categorical tracking bars.

An additional lollipop plot was generated to compare differences in mean gene expression between normal and preeclampsia pregnancies across the 10 genes included in the heatmap. Differences were calculated as mean expression among preeclampsia samples minus mean expression among normal pregnancy samples.

1.3 Software

All data cleaning, analysis, and visualization were performed using R version 4.6.1 [4]. The `tidyverse` package version 2.0.0 was used for data manipulation and visualization [5]. The `pheatmap` package version 1.0.13 was used to generate the clustered heatmap [1]. The `latex2exp` package version 0.9.8 was used to format gene names in plots [2].

The complete analysis code and project files are available in the project GitHub repository: `https://github.com/azairesandrehse-design/HSE103-Final-Project`

2 Results

2.1 Summary Statistics

The analytic sample included 110 participants with normal pregnancy outcomes and 47 participants with preeclampsia. Participants with preeclampsia had a higher mean maternal BMI and a lower mean maternal vitamin D concentration than participants with normal pregnancy outcomes (Table 1). Above-median BMI was also more common among participants with preeclampsia.

Table 1: Summary Statistics Stratified by Pregnancy Outcome

Variable	**Normal (n = 110)**	**Preeclampsia (n = 47)**
Maternal age, years	26.63 (5.08)	25.78 (4.95)
Maternal BMI	27.99 (7.66)	31.16 (8.03)
Maternal vitamin D, ng/mL	24.37 (14.55)	19.72 (8.33)
Maternal BMI group		
Below median BMI	61 (55.5%)	18 (38.3%)
Above median BMI	49 (44.5%)	29 (61.7%)
Maternal asthma		
Healthy	67 (60.9%)	27 (57.4%)
Asthmatic	43 (39.1%)	20 (42.6%)

Note. Continuous variables are presented as mean (standard deviation). Categorical variables are presented as n (%). BMI = body mass index.

2.2 Gene Expression Analyses

Among the three primary genes, *TBL1X* had the highest overall mean expression, followed by *TMEM176B* and *KDM1B*. The expression distributions for each gene are shown in Figures 1, 4, and 7.

The scatterplots show the observed relationships between maternal vitamin D concentration and expression of each of the three selected genes (Figures 2, 5, and 8). The fitted lines are presented descriptively to show the direction of the observed linear trend.

The grouped boxplots show the distributions of gene expression according to pregnancy outcome and maternal BMI group for *TBL1X*, *KDM1B*, and *TMEM176B* (Figures 3, 6, and 9).

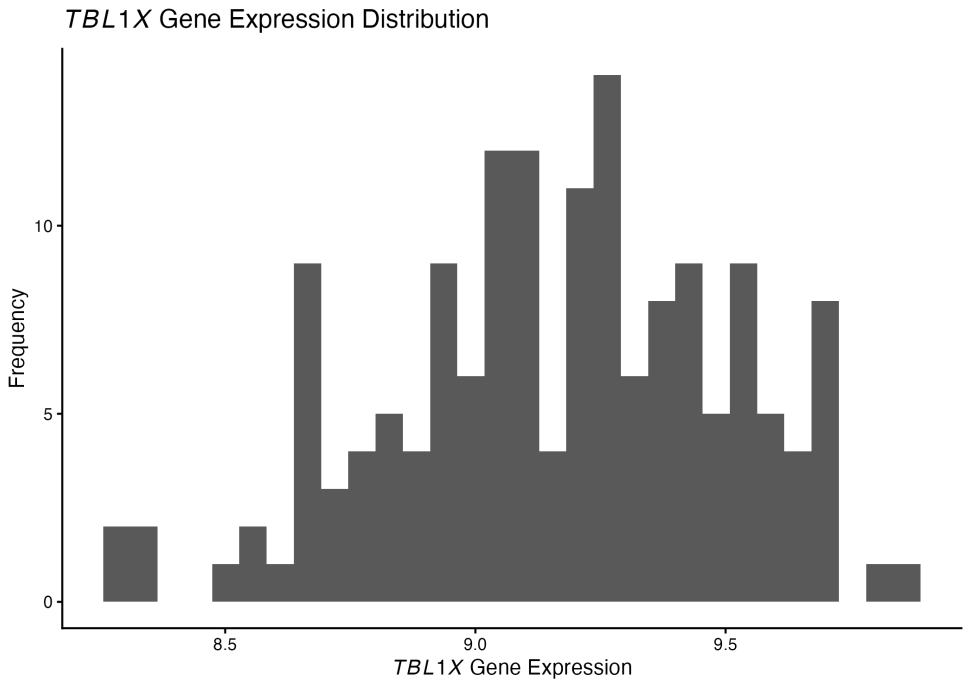


Figure 1: Distribution of *TBL1X* gene expression among the 157 maternal samples.

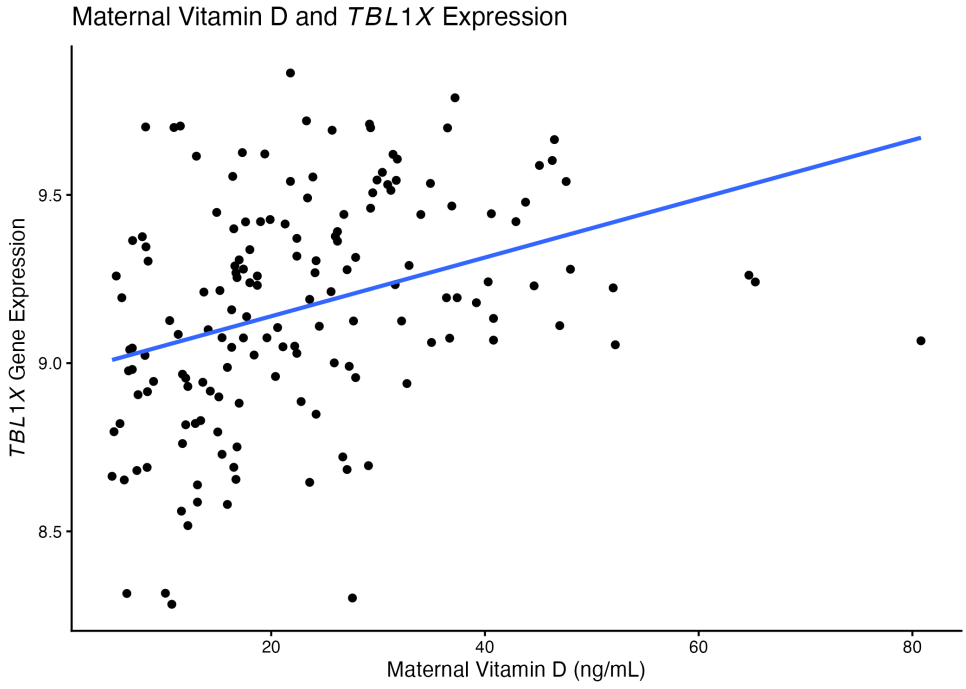


Figure 2: Relationship between maternal vitamin D concentration and *TBL1X* gene expression. The fitted line represents the observed linear trend.

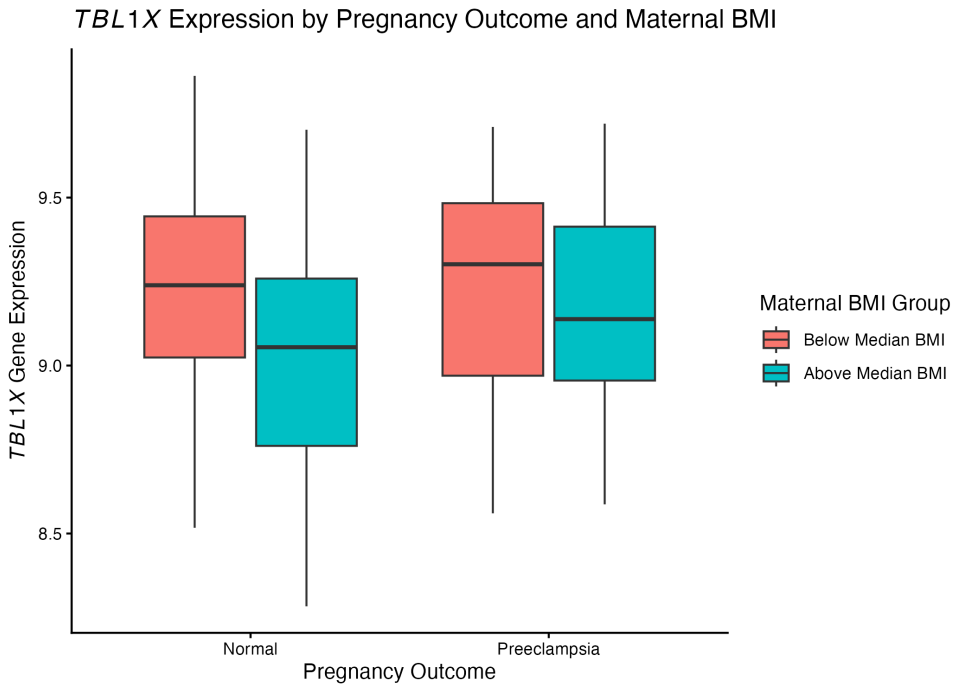


Figure 3: *TBL1X* gene expression according to pregnancy outcome and maternal BMI group.

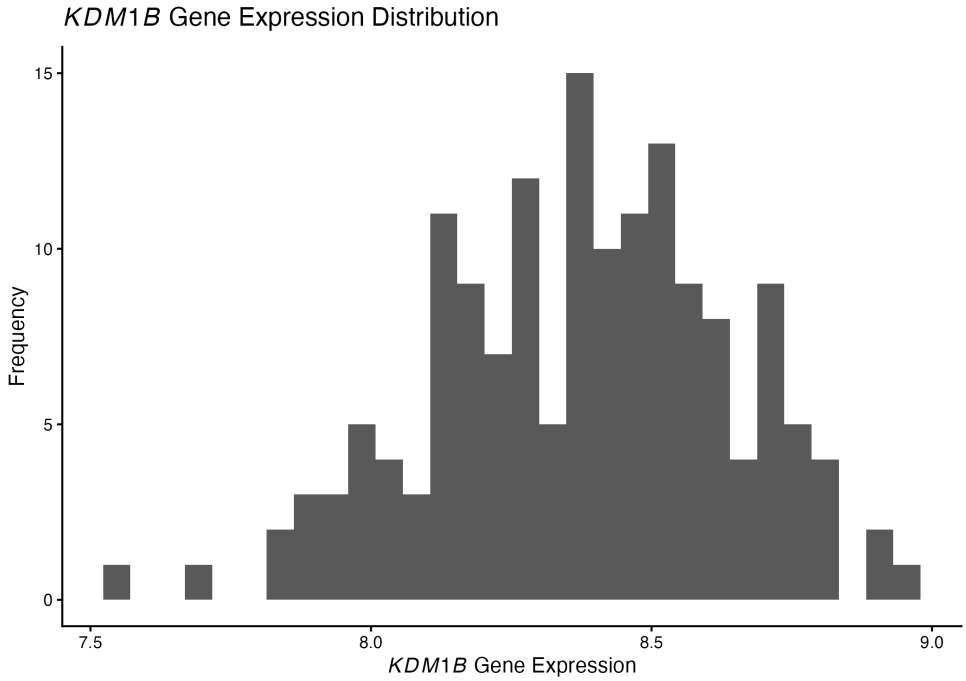


Figure 4: Distribution of *KDM1B* gene expression among the 157 maternal samples.

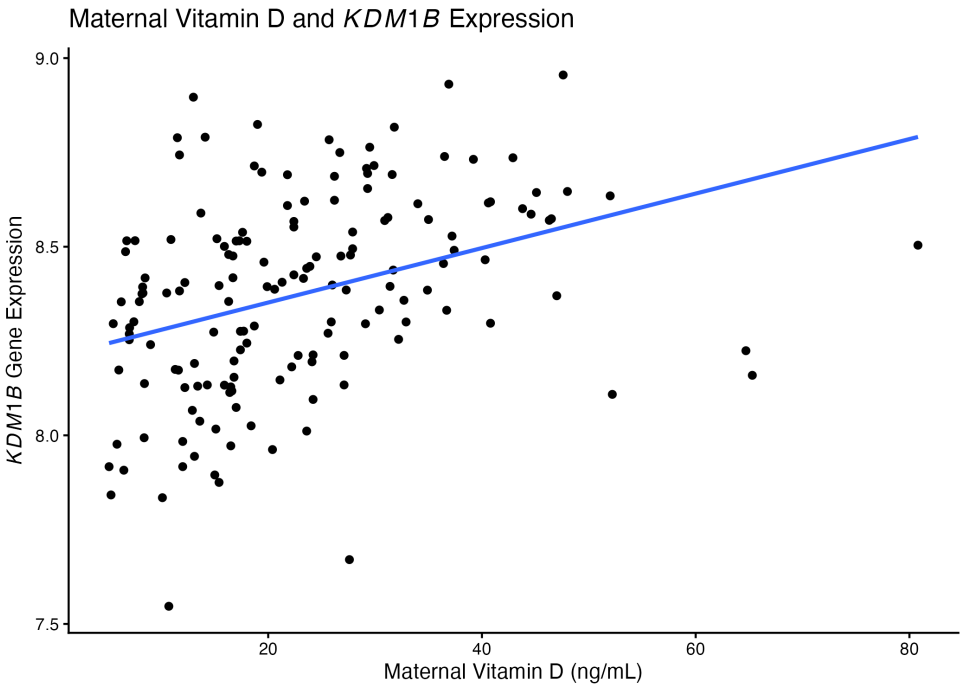


Figure 5: Relationship between maternal vitamin D concentration and *KDM1B* gene expression. The fitted line represents the observed linear trend.

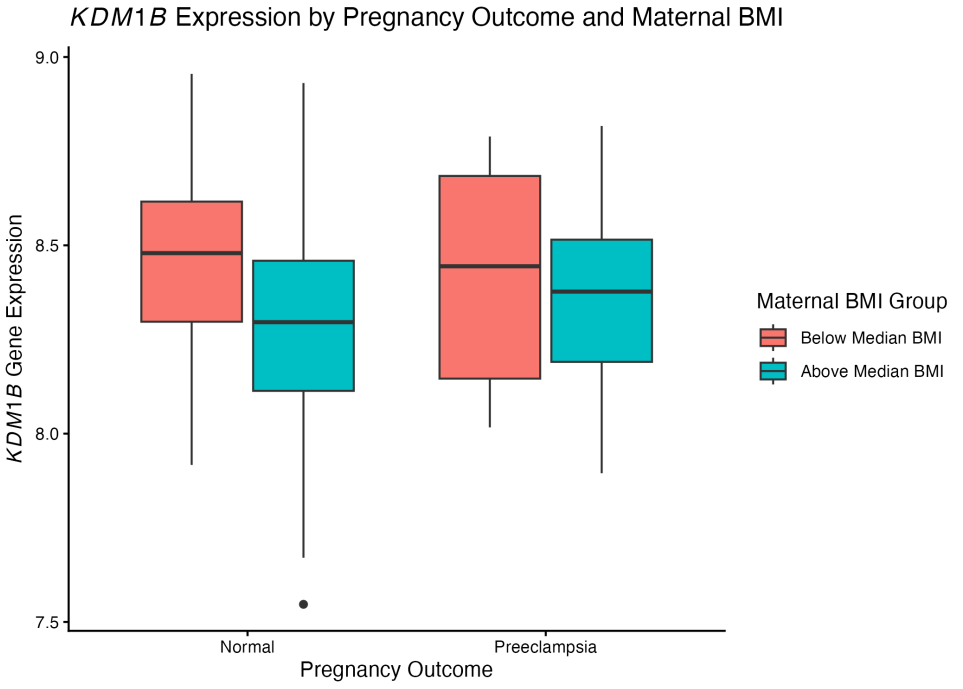


Figure 6: *KDM1B* gene expression according to pregnancy outcome and maternal BMI group.

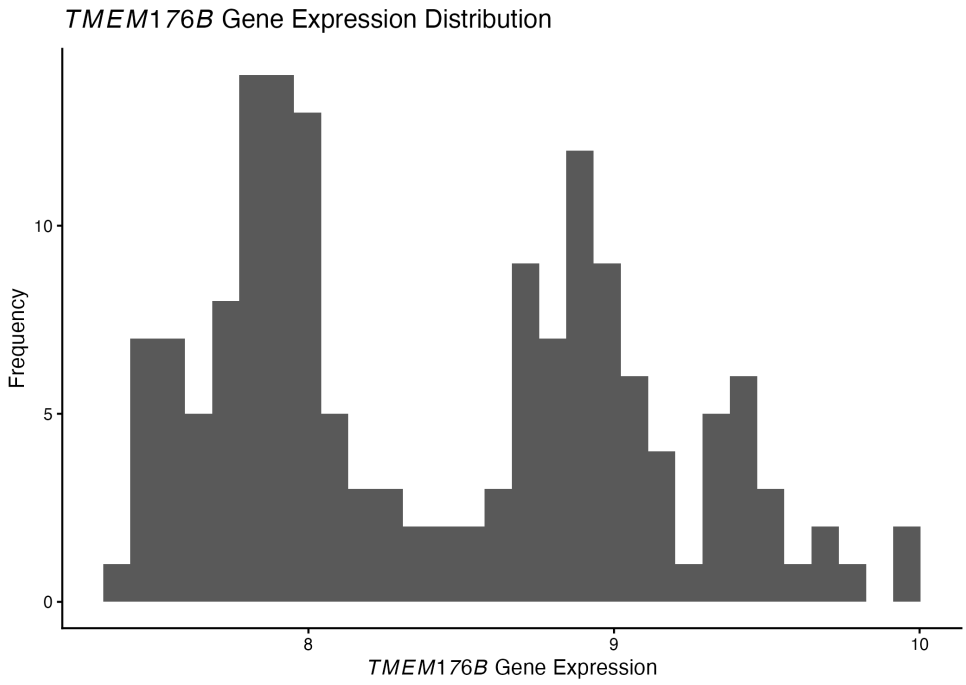


Figure 7: Distribution of *TMEM176B* gene expression among the 157 maternal samples.

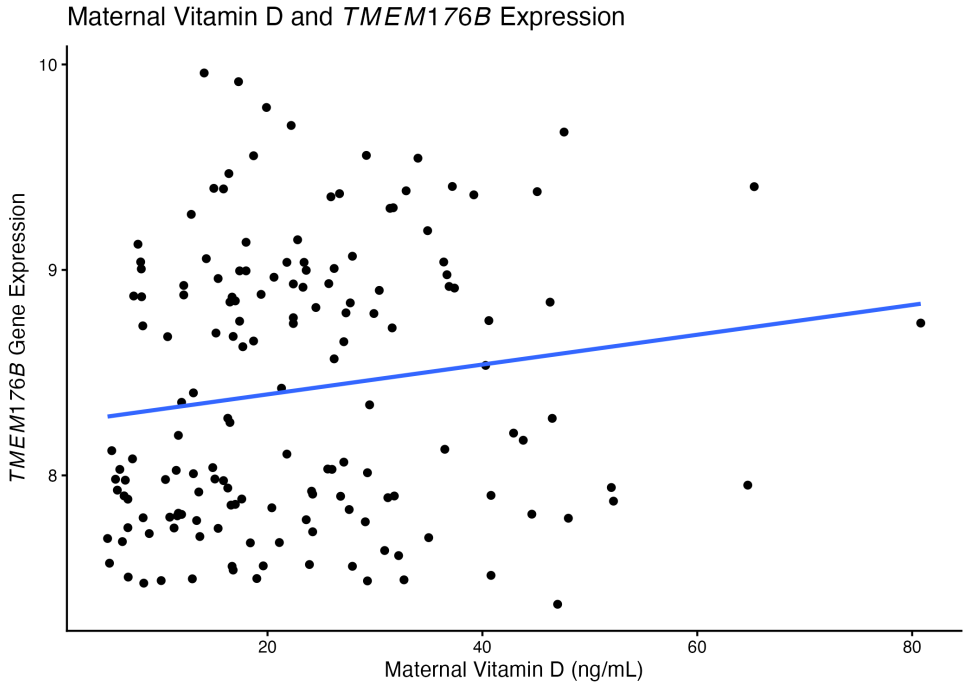


Figure 8: Relationship between maternal vitamin D concentration and *TMEM176B* gene expression. The fitted line represents the observed linear trend.

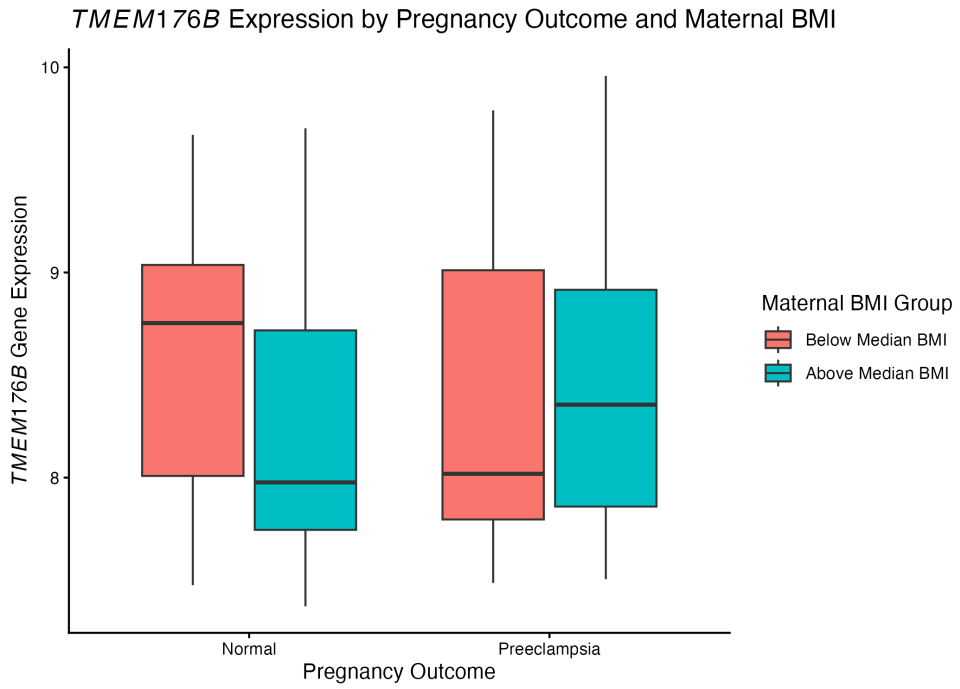


Figure 9: *TMEM176B* gene expression according to pregnancy outcome and maternal BMI group.

2.3 Heatmap

The clustered heatmap displays standardized expression patterns for the 10 selected genes across all 157 maternal samples (Figure 10). The genes included were *TBL1X*, *KDM1B*, *TMEM176B*, *UTS2*, *RUNDC3A*, *CEACAM6*, *CLIC2*, *SLC1A5*, *ZNF594*, and *ZBTB41*. Samples were grouped according to similarities in gene expression profiles, while the tracking bars show pregnancy outcome and maternal BMI group.

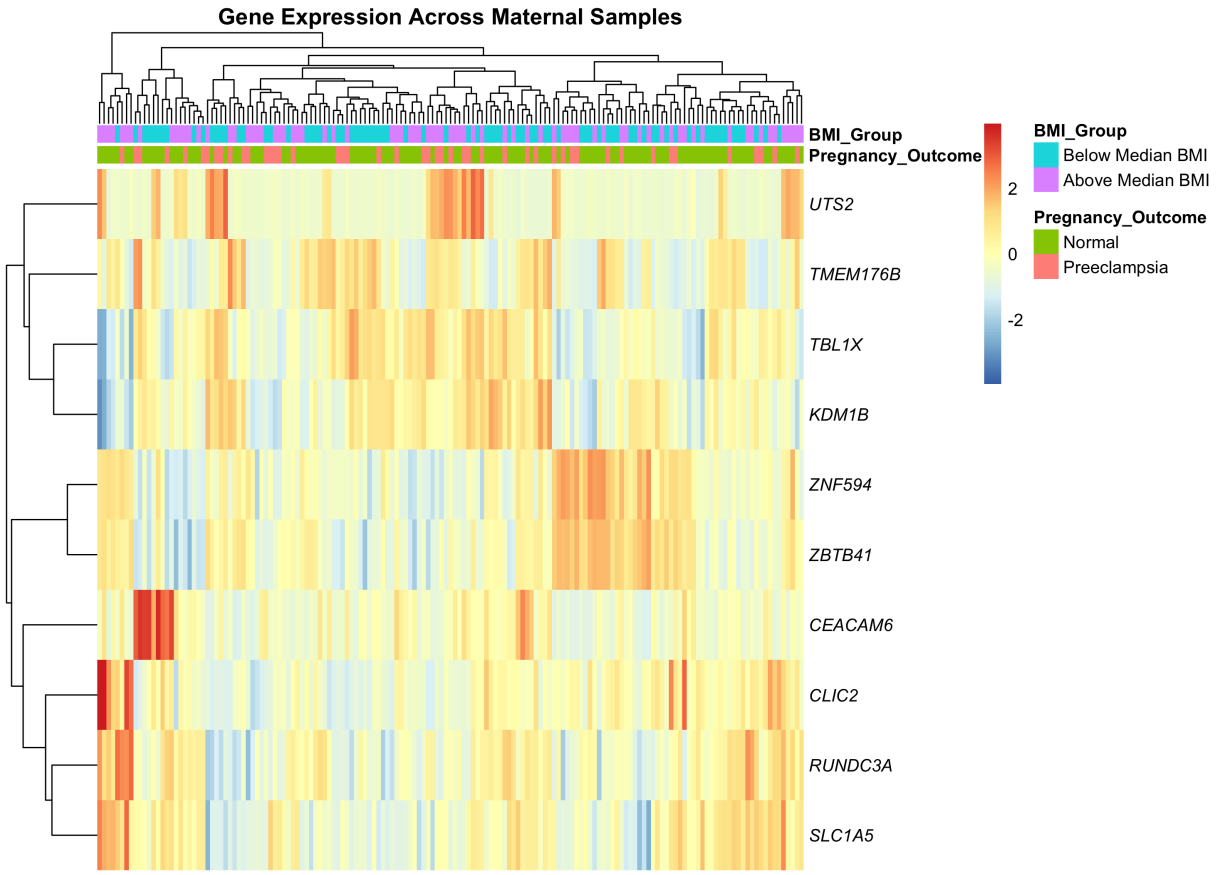


Figure 10: Clustered heatmap of standardized expression for 10 genes across 157 maternal samples. Rows represent genes and columns represent samples. Both genes and samples were hierarchically clustered using Euclidean distance. Tracking bars indicate pregnancy outcome and maternal BMI group.

2.4 Additional Visualization

The lollipop plot compares differences in mean gene expression between preeclampsia and normal pregnancy samples for the 10 genes included in the heatmap (Figure 11). Positive values represent higher mean expression in the preeclampsia group, while negative values represent higher mean expression in the normal pregnancy group.

Among the 10 genes, *UTS2* had the largest positive mean expression difference, approximately 0.217. *RUNDC3A* had the largest negative mean expression difference, approximately -0.261. *CLIC2* and *ZNF594* also showed lower mean expression in the preeclampsia group relative to the normal pregnancy group.

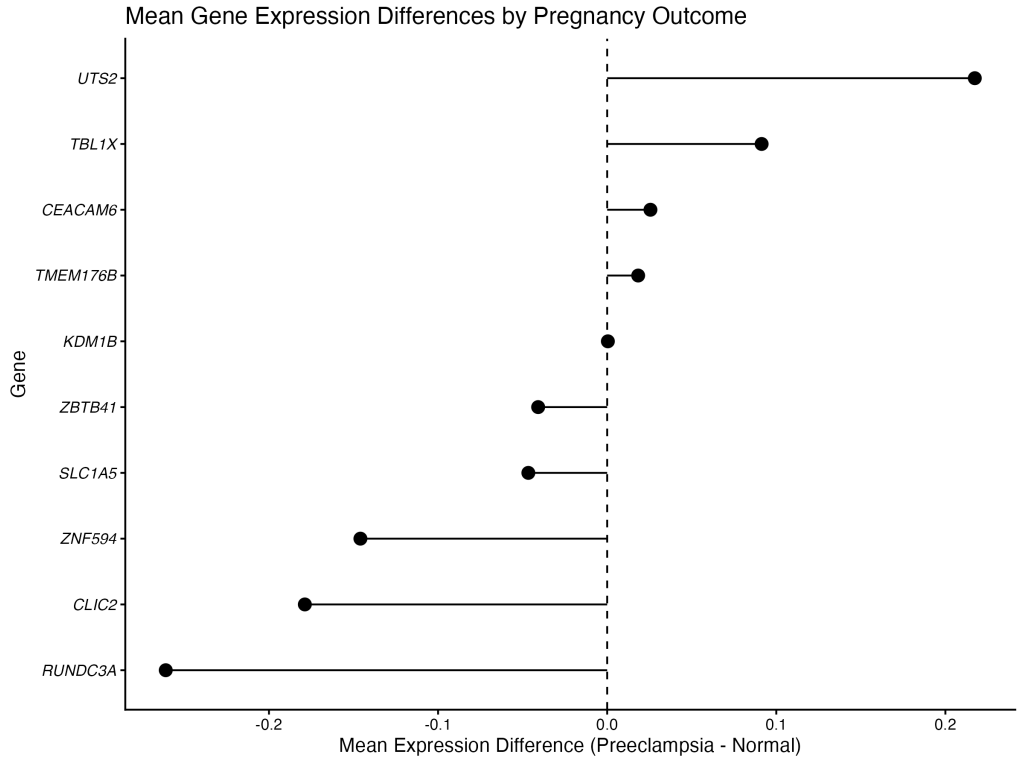


Figure 11: Difference in mean gene expression between preeclampsia and normal pregnancy samples for 10 selected genes. Values were calculated as mean expression in the preeclampsia group minus mean expression in the normal pregnancy group.

References

[1] Raivo Kolde. *pheatmap: Pretty Heatmaps*.

[2] Stefano Meschiari. *latex2exp: Use LaTeX Expressions in Plots*.

[3] Hooman Mirzakhani, Augusto A. Litonjua, Thomas F. McElrath, George O'Connor, Aviva Lee-Parritz, Ronald Iverson, George Macones, Robert C. Strunk, Leonard B. Bacharier, Robert Zeiger, Bruce W. Hollis, Diane E. Handy, Amitabh Sharma, Nancy Laranjo, Vincent Carey, Weiliang Qiu, Marc Santolini, Shikang Liu, Divya Chhabra, Daniel A. Enquobahrie, Michelle A. Williams, Joseph Loscalzo, and Scott T. Weiss. Early pregnancy vitamin d status and risk of preeclampsia. *The Journal of Clinical Investigation*, 126(12):4702–4715, 2016.

[4] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

[5] Hadley Wickham, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain Francois, Garrett Grolemund, Alex Hayes, Lionel Henry, Jim Hester, Max Kuhn, Thomas Lin Pedersen, Evan Miller, Stephan Milton Bache, Kirill Muller, Jeroen Ooms, David Robinson, Dana Paige Seidel, Vitalie Spinu, Kohske Takahashi, Davis Vaughan, Claus Wilke, Kara Woo, and Hiroaki Yutani. Welcome to the tidyverse. *Journal of Open Source Software*, 4(43):1686, 2019.